



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

General Chemistry CHM1001

Nuclear Chemistry (Ch. 21)

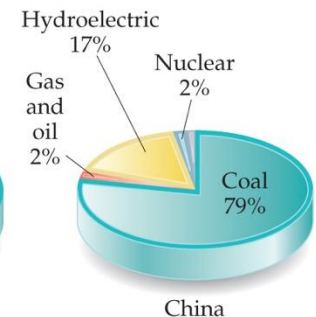
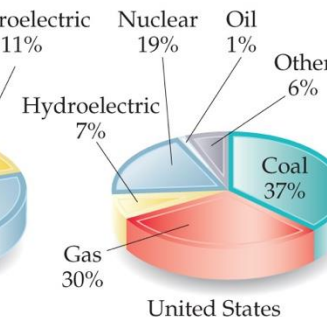
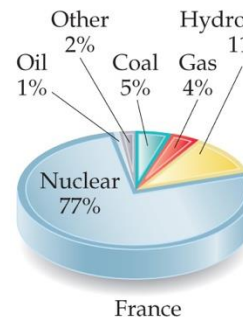
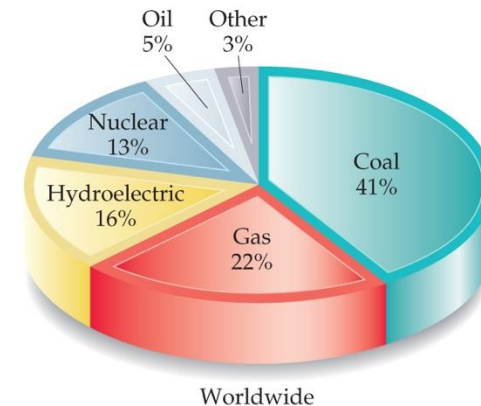
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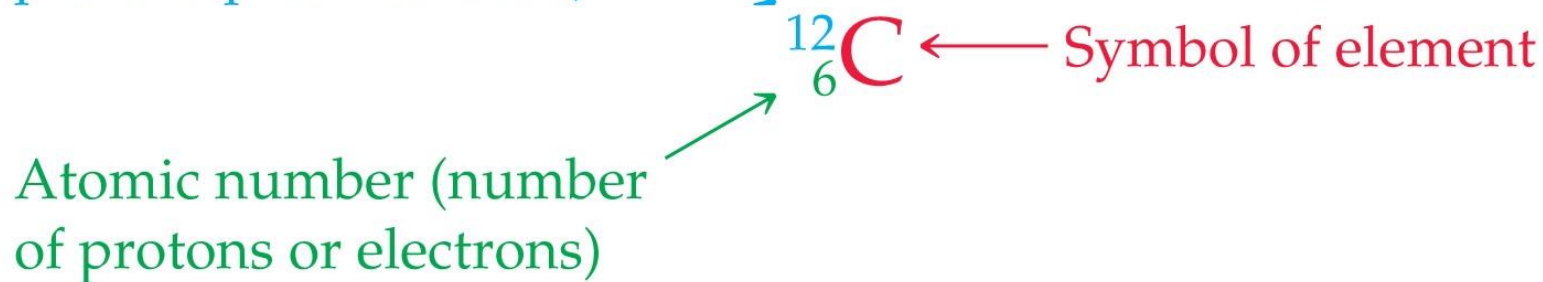
Energy: Chemical vs. Nuclear

- Chemical energy is associated with making and breaking chemical bonds.
- Nuclear energy is enormous in comparison.
- Nuclear energy is due to changes in the nucleus of atoms changing them into different atoms.
- 13% of worldwide energy use comes from nuclear energy.



The Nucleus

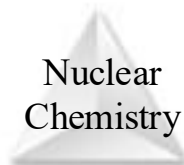
Mass number (number of protons plus neutrons)



- Remember that the nucleus is composed of the **two nucleons, protons and neutrons.**
- The number of protons is the atomic number.
- The number of protons and neutrons together is the mass number.

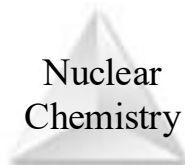
Isotopes

- Not all atoms of the same element have the same mass, due to different numbers of neutrons in those atoms.
- There are, for example, three naturally occurring isotopes of uranium:
 - Uranium-234
 - Uranium-235
 - Uranium-238



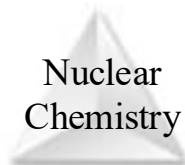
Radioactivity

- It is not uncommon for some **nuclides** of an element to be **unstable**, or **radioactive**.
- We refer to these as **radionuclides**.
- There are several ways radionuclides can decay into a different nuclide.
- We use nuclear equations to show how these nuclear reactions occur.



Equations

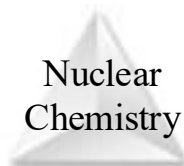
- In chemical equations, atoms and charges need to balance.
- In nuclear equations, atomic number and mass number need to balance. This is a way of balancing charge (atomic number) and mass (mass number) on an atomic scale.



Most Common Kinds of Radiation Emitted by a Radionuclide

Table 21.1 Properties of Alpha, Beta, and Gamma Radiation

Property	Type of Radiation		
	α	β	γ
Charge	2+	1−	0
Mass	$6.64 \times 10^{-24}\text{g}$	$9.11 \times 10^{-28}\text{g}$	0
Relative penetrating power	1	100	10,000
Nature of radiation	${}^4_2\text{He}$ nuclei	Electrons	High-energy photons



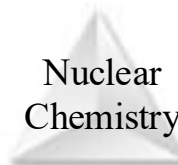
Types of Radioactive Decay

- Alpha decay
- Beta decay
- Gamma emission
- Positron emission
- Electron capture

Table 21.3 Types of Radioactive Decay

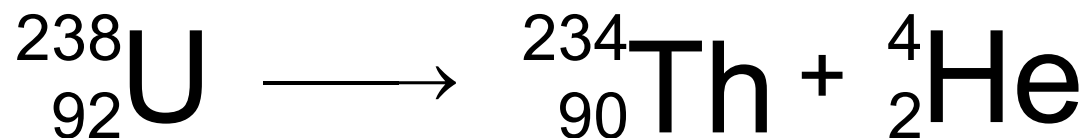
Type	Nuclear Equation	Change in Atomic Number	Change in Mass Number
Alpha decay	${}^A_Z\text{X} \longrightarrow {}^{A-4}_{Z-2}\text{Y} + {}^4_2\text{He}$	-2	-4
Beta emission	${}^A_Z\text{X} \longrightarrow {}^A_{Z+1}\text{Y} + {}^0_{-1}\text{e}$	+1	Unchanged
Positron emission	${}^A_Z\text{X} \longrightarrow {}^A_{Z-1}\text{Y} + {}^0_{+1}\text{e}$	-1	Unchanged
Electron capture*	${}^A_Z\text{X} + {}^0_{-1}\text{e} \longrightarrow {}^A_{Z-1}\text{Y}$	-1	Unchanged

*The electron captured comes from the electron cloud surrounding the nucleus.



Alpha Decay

Alpha decay is the loss of an α -particle (He-4 nucleus, two protons and two neutrons):

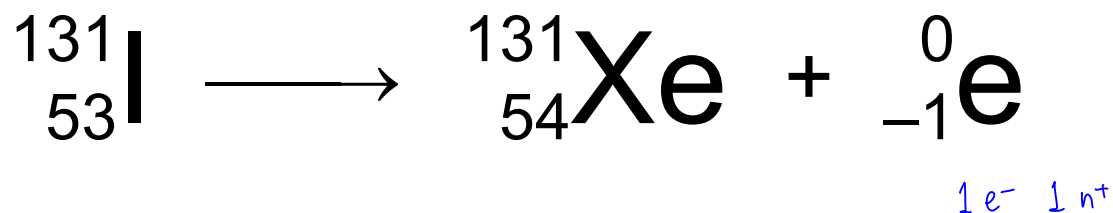
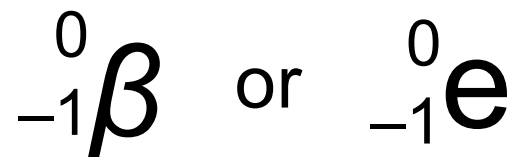


Note how the equation balances:

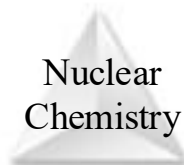
- atomic number: $92 = 90 + 2$
- mass number: $238 = 234 + 4$

Beta Decay

Beta decay is the loss of a β -particle (a high-speed electron emitted by the nucleus):

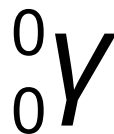


Balancing: atomic number: $53 = 54 + (-1)$
mass number: $131 = 131 + 0$



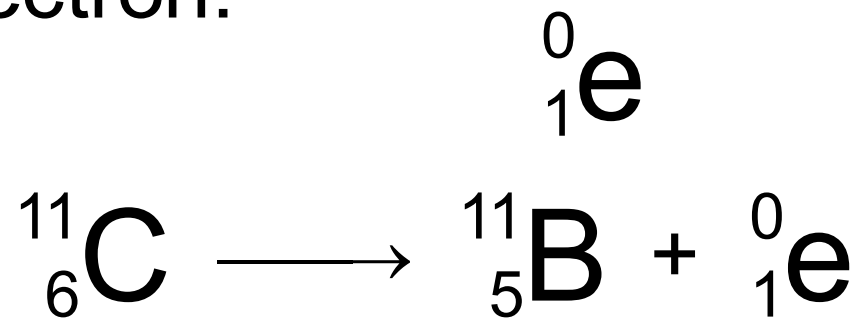
Gamma Emission

Gamma emission is the loss of a γ -ray, which is high-energy radiation that almost always accompanies the loss of a nuclear particle:

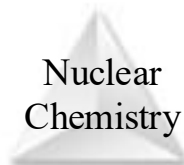


Positron Emission

Some nuclei decay by emitting a **positron**, a particle that has the same mass as, but an opposite charge to, that of an electron:

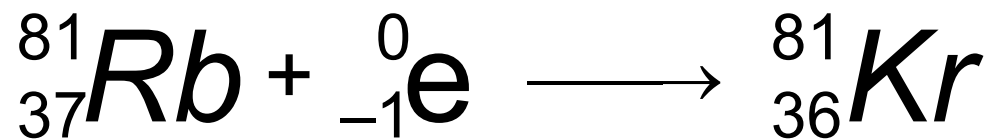


Balancing: atomic number: $6 = 5 + 1$
mass number: $11 = 11 + 0$



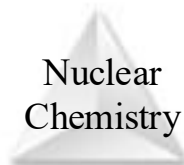
Electron Capture (K-Capture)

An electron from the surrounding electron cloud is absorbed into the nucleus during **electron capture**.



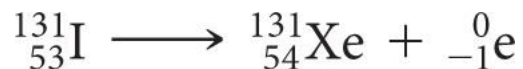
O^{2-} nucleus did not change

Balancing: atomic number: $37 + (-1) = 36$
mass number: $81 + 0 = 81$



Sources of Some Nuclear Particles

- Beta particles:



- Positrons: (p = proton)



- What happens with electron capture?

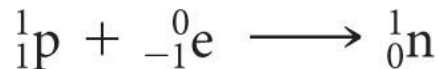
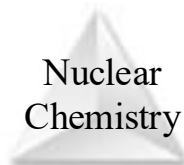


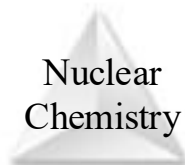
Table 21.2 Particles Found in Nuclear Reactions

Particle	Symbol
Neutron	${}_0^1\text{n}$ or n
Proton	${}_1^1\text{H}$ or p
Electron	${}_{-1}^0\text{e}$
Alpha particle	${}_2^4\text{He}$ or α
Beta particle	${}_{-1}^0\text{e}$ or β^-
Positron	${}_{+1}^0\text{e}$ or β^+



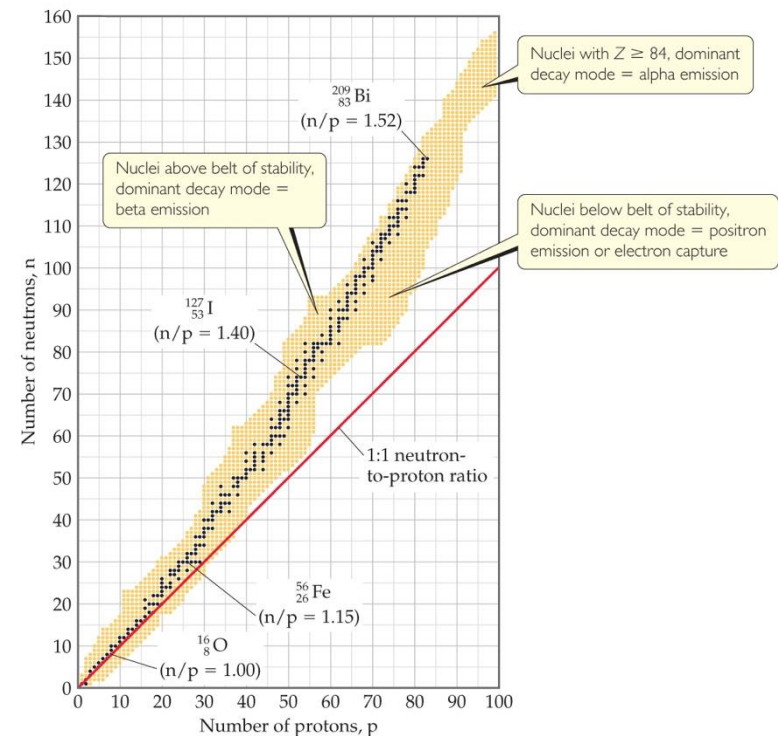
Nuclear Stability

- Any atom with more than one proton (anything but H) will have repulsions between the protons in the nucleus.
- **Strong nuclear force** helps keep the nucleus together.
- Neutrons play a key role stabilizing the nucleus, so the ratio of neutrons to protons is an important factor.



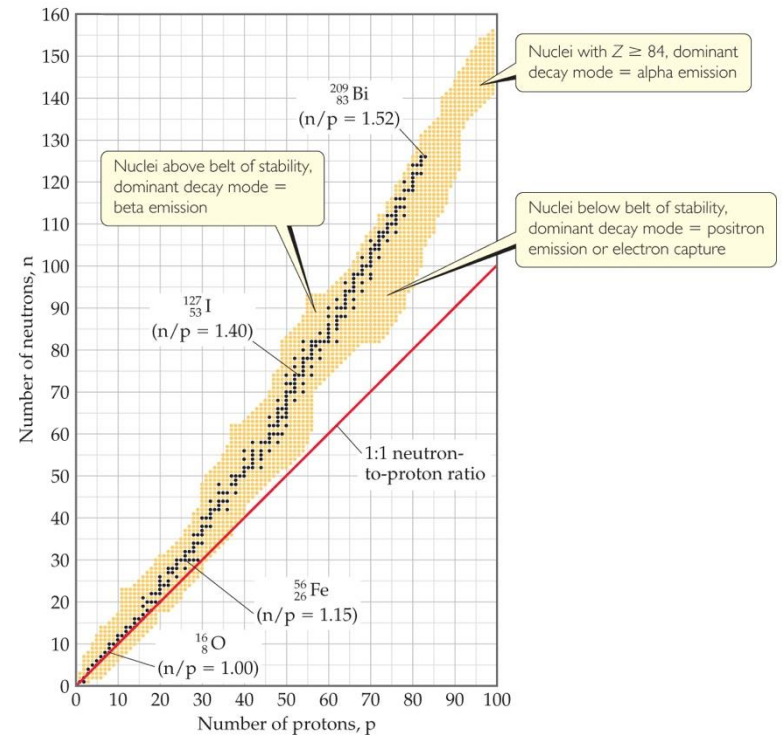
Neutron–Proton Ratios

- For smaller nuclei ($Z \leq 20$), stable nuclei have a neutron-to-proton ratio close to 1:1.
- As nuclei get larger, it takes a larger number of neutrons to stabilize the nucleus.
- The shaded region in the figure is called the **belt of stability**; it shows what nuclides would be stable.



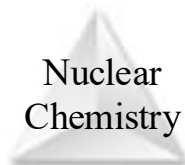
Unstable Nuclei

- Compare a nucleus to the “belt of stability.”
- Nuclei above this belt have too many neutrons, so they tend to decay by emitting beta particles.
- Nuclei below the belt have too many protons, so they tend to become more stable by positron emission or electron capture.



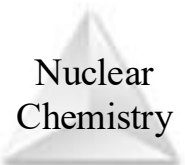
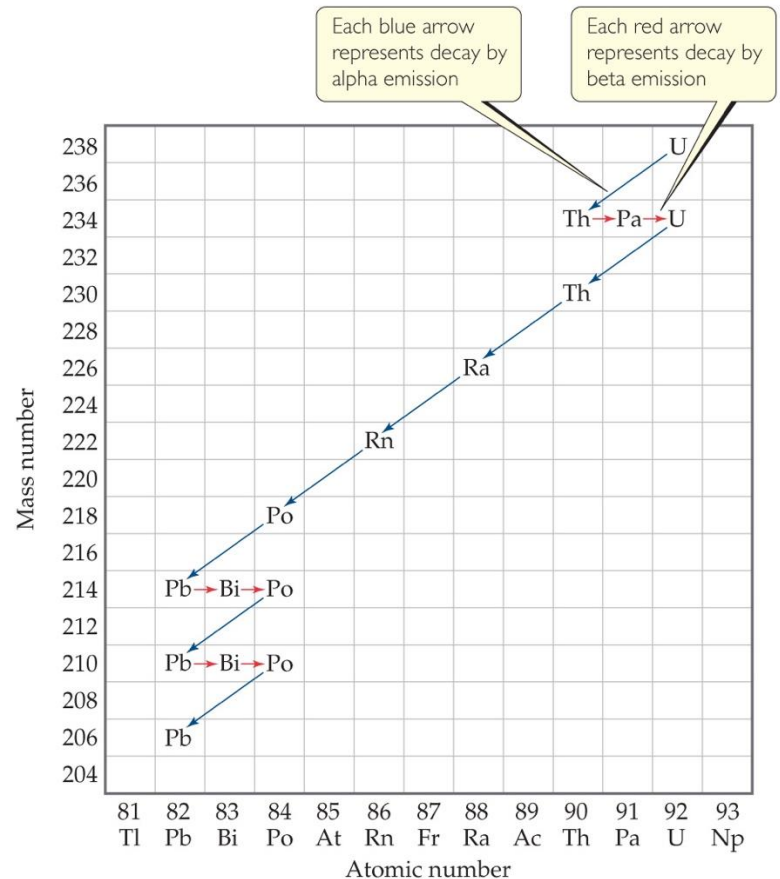
Alpha Emission

- There are no stable nuclei with an atomic number greater than 83.
- Nuclei with such large atomic numbers tend to decay by alpha emission.



Radioactive Decay Chain

- Some radioactive nuclei cannot stabilize by undergoing only one nuclear transformation.
- They undergo a series of decays until they form a **stable nuclide** (often a nuclide of **lead**).



Stable Nuclei

1 H (2)																		2 He (2)																	
3 Li (2)		4 Be (1)																		5 B (2)		6 C (2)		7 N (2)		8 O (3)		9 F (1)		10 Ne (3)					
11 Na (1)		12 Mg (3)																		13 Al (1)		14 Si (3)		15 P (1)		16 S (4)		17 Cl (2)		18 Ar (3)					
19 K (2)		20 Ca (5)		21 Sc (1)		22 Ti (5)		23 V (2)		24 Cr (4)		25 Mn (1)		26 Fe (4)		27 Co (1)		28 Ni (5)		29 Cu (2)		30 Zn (5)		31 Ga (2)		32 Ge (4)		33 As (1)		34 Se (5)		35 Br (2)		36 Kr (6)	
37 Rb (1)		38 Sr (3)		39 Y (1)		40 Zr (4)		41 Nb (1)		42 Mo (6)		43 Tc (0)		44 Ru (7)		45 Rh (1)		46 Pd (6)		47 Ag (2)		48 Cd (6)		49 In (1)		50 Sn (10)		51 Sb (2)		52 Te (6)		53 I (1)		54 Xe (9)	

- **Magic numbers of 2, 8, 20, 28, 50, or 82 protons or 2, 8, 20, 28, 50, 82, or 126 neutrons result in more stable nuclides.**
- Nuclei with an **even number of protons and neutrons tend to be more stable** than those with odd numbers.

Table 21.4 Number of Stable Isotopes with Even and Odd Numbers of Protons and Neutrons

Number of Stable Isotopes	Proton Number	Neutron Number
157	Even	Even
53	Even	Odd
50	Odd	Even
5	Odd	Odd

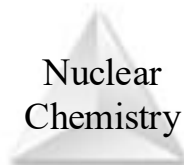
Nuclear Transmutations

- **Nuclear transmutations** can be induced by accelerating a particle to collide it with the nuclide.
- **Particle accelerators** (“atom smashers”) are enormous, having circular tracks with radii that are miles long.



Other Nuclear Transmutations

- Use of neutrons:
 - Most synthetic isotopes used in medicine are prepared by bombarding neutrons at a particle, which won't repel the neutral particle.
- Transuranium elements:
 - Elements immediately after uranium were discovered by bombarding isotopes with neutrons.
 - Larger elements (atomic number higher than 110) were made by colliding large atoms with nuclei of light elements with high energy.

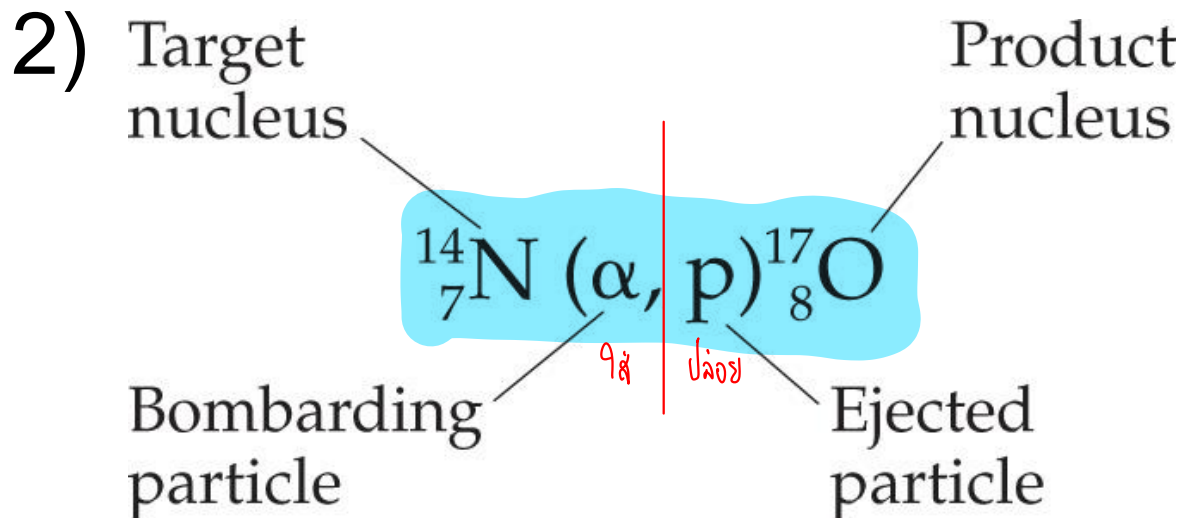


Writing Nuclear Equations for Nuclear Transmutations

Nuclear equations that represent nuclear transmutations are written two ways:



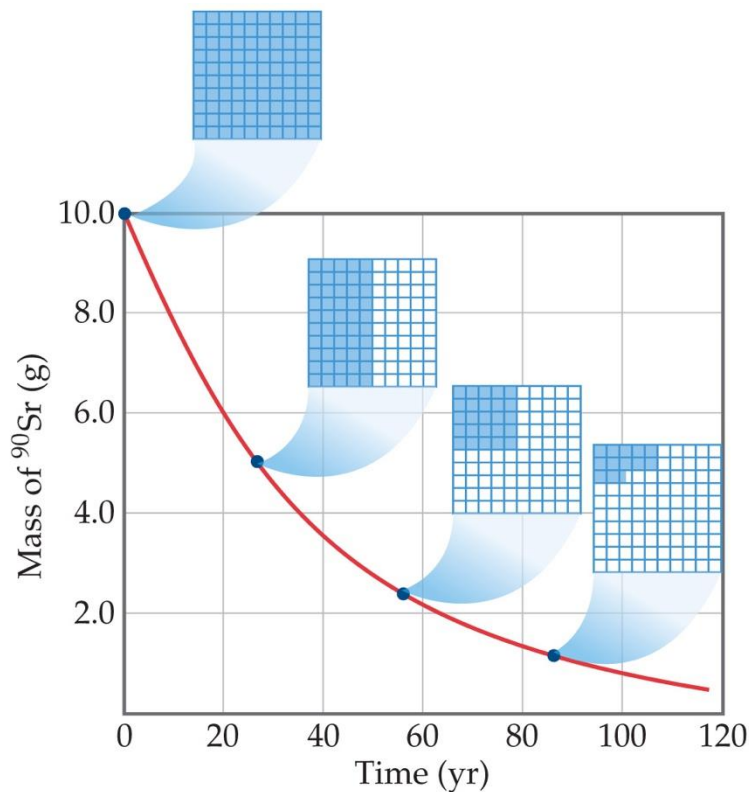
or



Kinetics of Radioactive Decay

- Radioactive decay is a first-order process.
- The kinetics of such a process obey this equation:

$$\ln \frac{N_t}{N_0} = -kt$$



Half-Life

- The half-life of such a process is

$$\frac{0.693}{k} = t_{1/2}$$

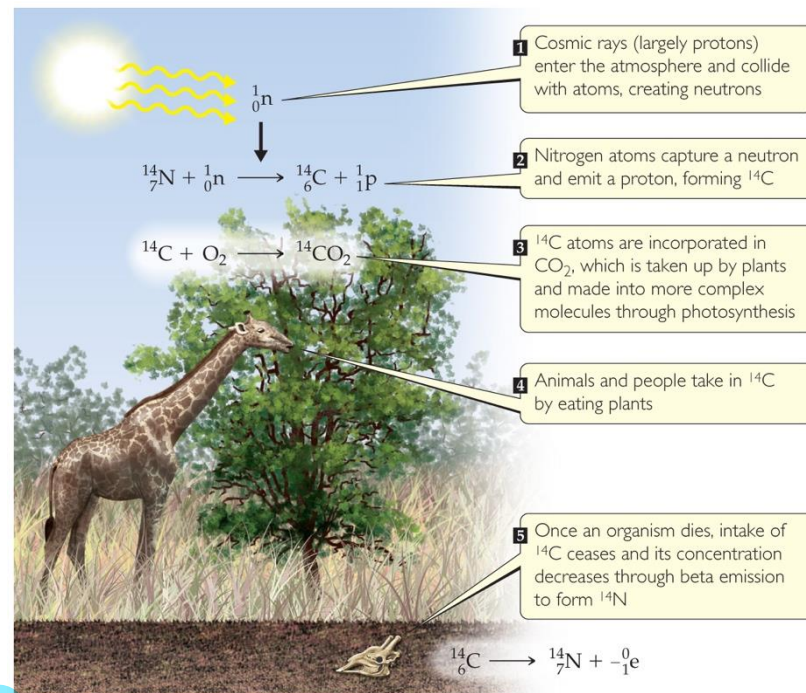
- Half-life is the time required for half of a radionuclide sample to decay.

Table 21.5 The Half-Lives and Type of Decay for Several Radioisotopes

	Isotope	Half-Life (yr)	Type of Decay
Natural radioisotopes	$^{238}_{92}\text{U}$	4.5×10^9	Alpha
	$^{235}_{92}\text{U}$	7.0×10^8	Alpha
	$^{232}_{90}\text{Th}$	1.4×10^{10}	Alpha
	$^{40}_{19}\text{K}$	1.3×10^9	Beta
	$^{14}_6\text{C}$	5700	Beta
Synthetic radioisotopes	$^{239}_{94}\text{Pu}$	24,000	Alpha
	$^{137}_{55}\text{Cs}$	30.2	Beta
	$^{90}_{38}\text{Sr}$	28.8	Beta
	$^{131}_{53}\text{I}$	0.022	Beta

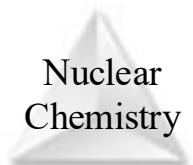
Radiometric Dating

- Applying first-order kinetics and half-life information, we can date objects using a “nuclear clock.”
- Carbon dating works: the half-life of C-14 is 5700 yr. It is limited to objects up to about 50,000 yr old; after this time there is too little radioactivity left to measure.
- Other isotopes can be used (U-238:Pb-206 in rock).



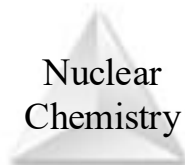
Measuring Radioactivity: Units

- **Activity** is the rate at which a sample decays.
- The units used to measure activity are as follows:
 - **Becquerel (Bq)**: one disintegration per second
 - **Curie (Ci)**: 3.7×10^{10} disintegrations per second, which is the rate of decay of 1 g of radium.



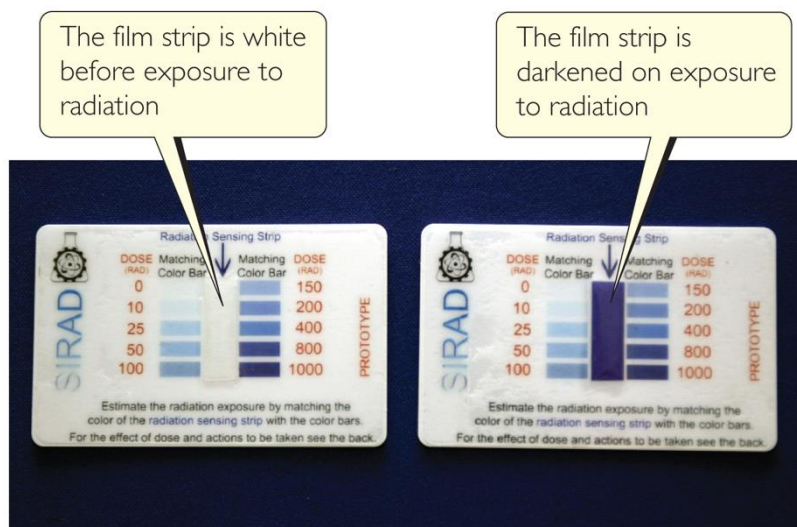
Measuring Radioactivity: Some Instruments

- Film badges
- Geiger counter
- Phosphors (scintillation counters)



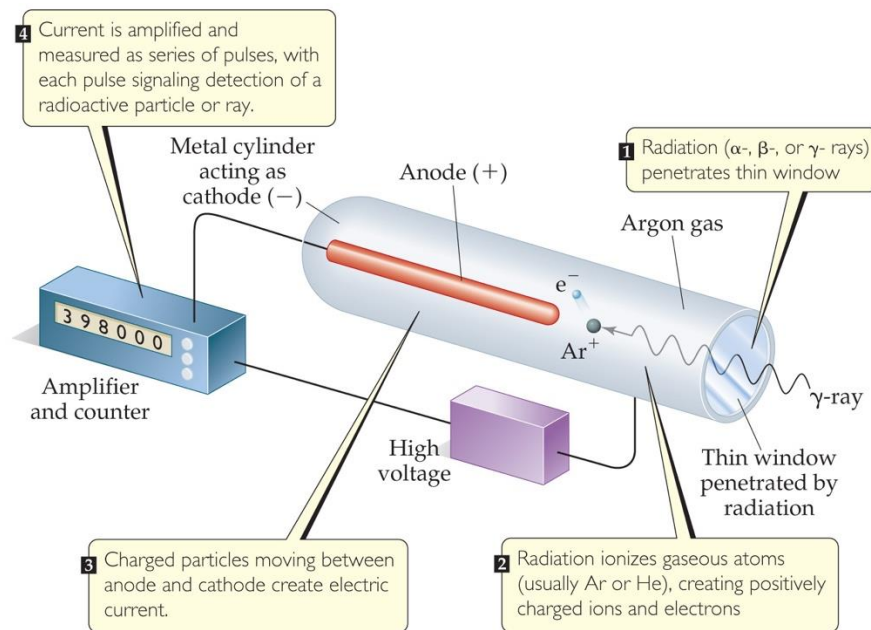
Film Badges

- Radioactivity was first discovered by Henri Becquerel because it fogged up a photographic plate.
- Film has been used to detect radioactivity since more exposure to radioactivity means darker spots on the developed film.
- Film badges are used by people who work with radioactivity to measure their own exposure over time.



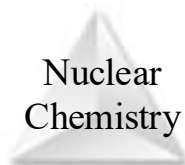
Geiger Counter

- A **Geiger counter** measures the amount of activity present in a radioactive sample.
- Radioactivity enters a window and creates ions in a gas; the ions result in an electric current that is measured and recorded by the instrument.



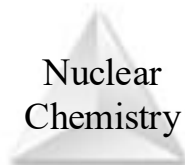
Phosphors

- Some substances absorb radioactivity and emit light. They are called **phosphors**.
- An instrument commonly used to measure the amount of light emitted by a phosphor is a scintillation counter. It converts the light to an electronic response for measurement.



Radiotracers

- **Radiotracers** are radioisotopes used to study a chemical reaction.
- An element can be followed through a reaction to determine its path and better understand the mechanism of a chemical reaction.



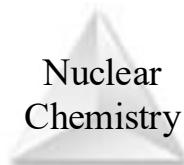
Medical Application of Radiotracers

- Radiotracers have found wide diagnostic use in medicine.
- Radioisotopes are administered to a patient (usually intravenously) and followed. Certain elements collect more in certain tissues, so an organ or tissue type can be studied based on where the radioactivity collects.

Table 21.6 Some Radionuclides Used as Radiotracers

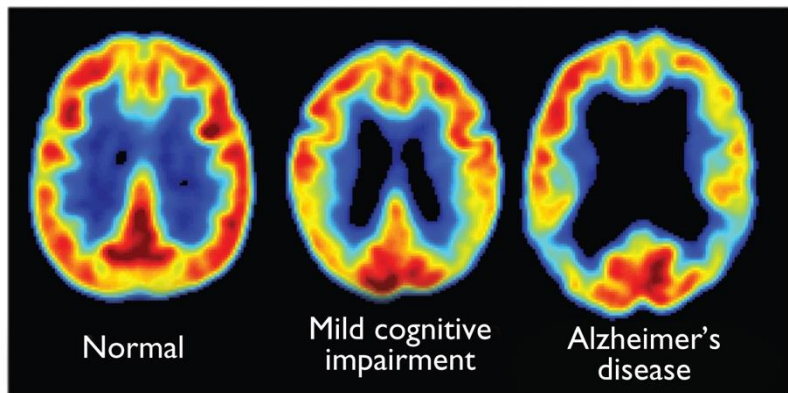
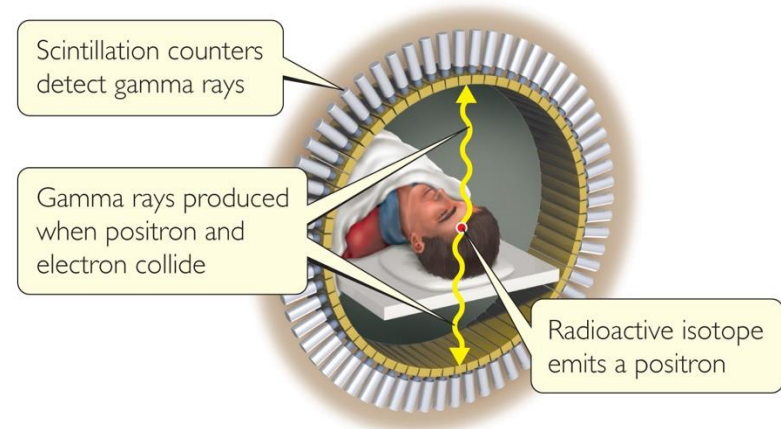
Nuclide	Half-Life	Area of the Body Studied
Iodine-131	8.04 days	Thyroid
Iron-59	44.5 days	Red blood cells
Phosphorus-32	14.3 days	Eyes, liver, tumors
Technetium-99 ^a	6.0 hours	Heart, bones, liver, and lungs
Thallium-201	73 hours	Heart, arteries
Sodium-24	14.8 hours	Circulatory system

^aThe isotope of technetium is actually a special isotope of Tc-99 called Tc-99m, where the *m* indicates a so-called *metastable* isotope.



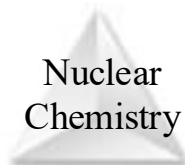
Positron Emission Tomography (PET Scan)

- A compound labeled with a positron emitter is injected into a patient.
- Blood flow, oxygen and glucose metabolism, and other biological functions can be studied.
- Labeled glucose is used to study the brain, as seen in the figure to the right.



Energy in Nuclear Reactions

- There is a tremendous amount of energy stored in nuclei.
- Einstein's famous equation, $E = mc^2$, relates directly to the calculation of this energy.



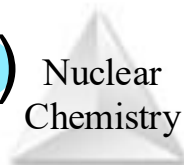
Energy in Nuclear Reactions

To show the enormous difference in energy for nuclear reactions, the mass change associated with the α -decay of 1 mol of U-238 to Th-234 is -0.0046 g.

The change in energy, ΔE , is then

$$\begin{aligned}\Delta E &= (\Delta m)c^2 \\ &= (-4.6 \times 10^{-6} \text{ kg})(3.00 \times 10^8 \text{ m/s})^2 \\ &= -4.1 \times 10^{11} \text{ J}\end{aligned}$$

(Note: the negative sign means heat is released.)

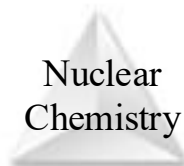


Mass Defect

- Where does this energy come from?
- The masses of nuclei are always less than those of the individual parts.
- This mass difference is called the **mass defect**.
- The energy needed to separate a nucleus into its nucleons is called the **nuclear binding energy**.

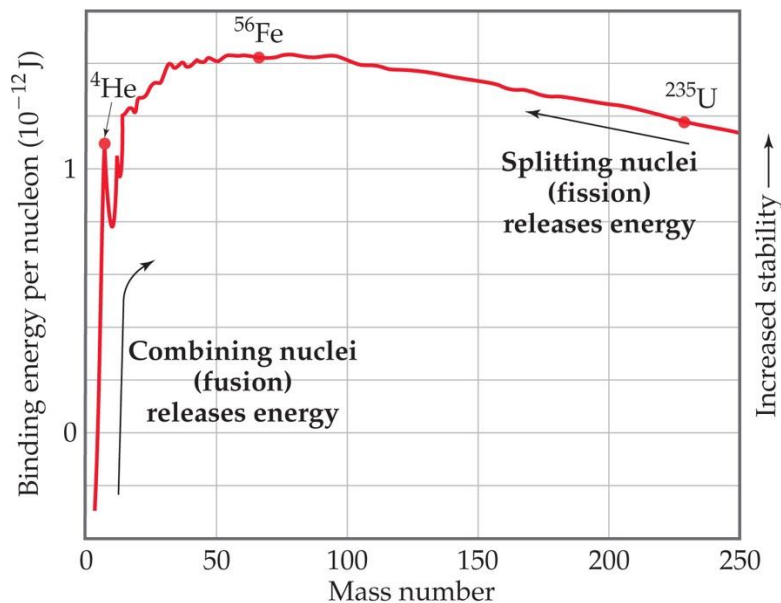
Table 21.7 Mass Defects and Binding Energies for Three Nuclei

Nucleus	Mass of Nucleus (amu)	Mass of Individual Nucleons (amu)	Mass Defect (amu)	Binding Energy (J)	Binding Energy per Nucleon (J)
${}^4_2\text{He}$	4.00150	4.03188	0.03038	4.53×10^{-12}	1.13×10^{-12}
${}^{56}_{26}\text{Fe}$	55.92068	56.44914	0.52846	7.90×10^{-11}	1.41×10^{-12}
${}^{238}_{92}\text{U}$	238.00031	239.93451	1.93420	2.89×10^{-10}	1.21×10^{-12}



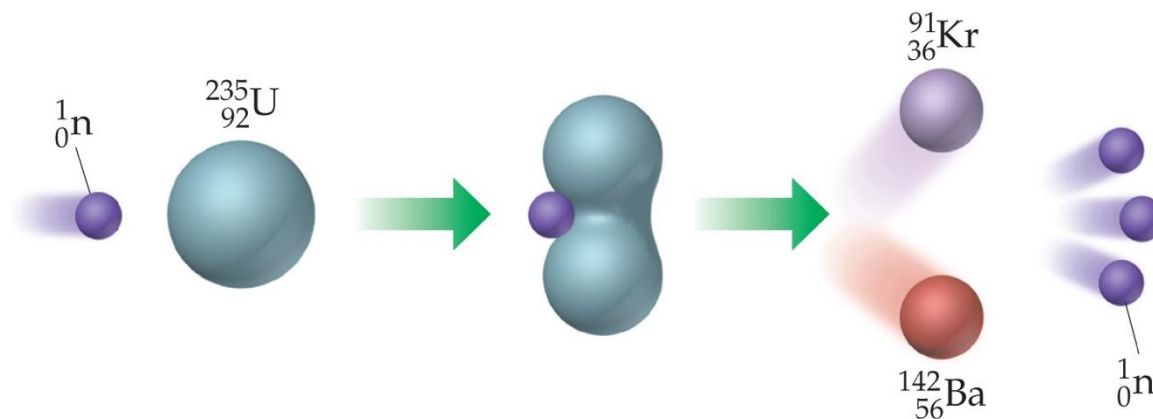
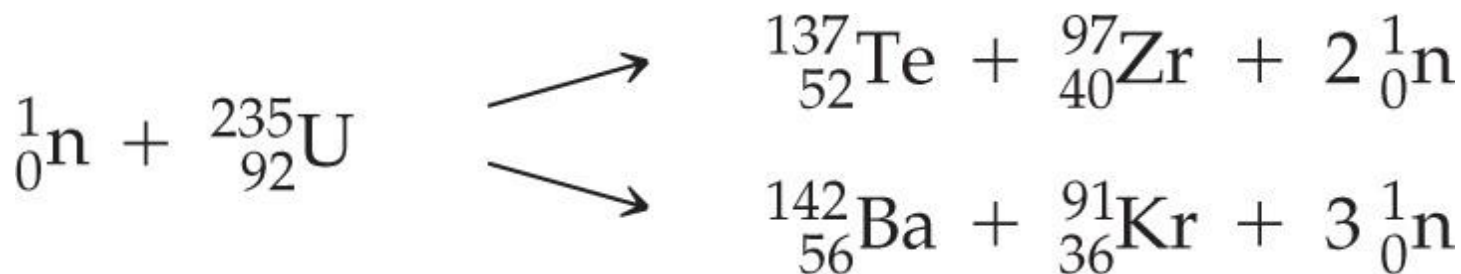
Effects of Nuclear Binding Energy on Nuclear Processes

- Dividing the binding energy by the number of nucleons gives a value that can be compared.
- Heavy nuclei gain stability and give off energy when they split into two smaller nuclei. This is **fission**.
- Lighter nuclei emit great amounts of energy by being combined in **fusion**.

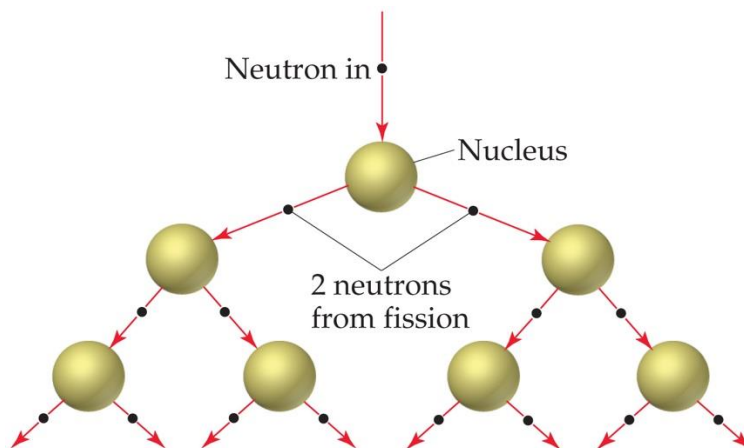


Nuclear Fission

- Commercial nuclear power plants use fission.
- Heavy nuclei can split in many ways. The equations below show two ways U-235 can split after bombardment with a neutron.

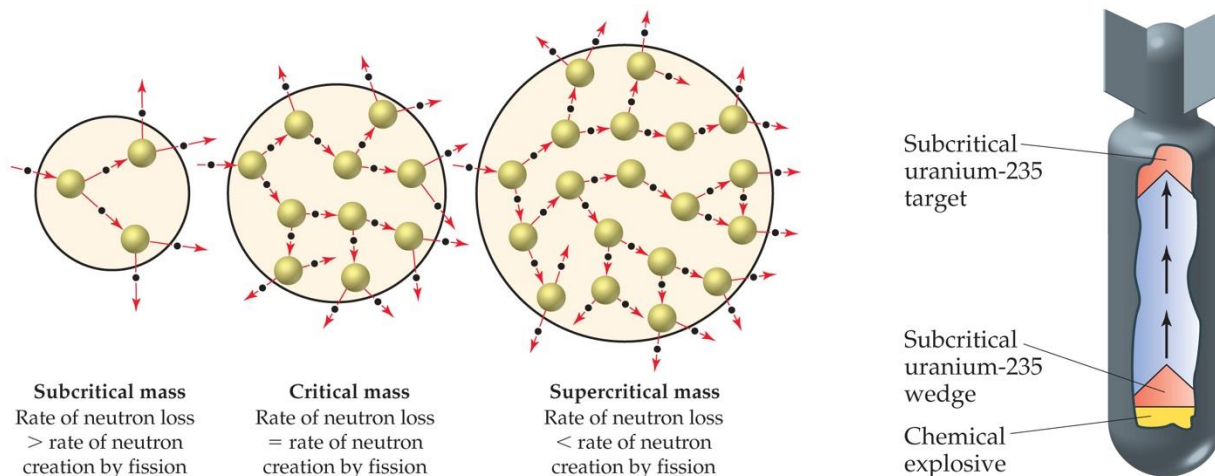


Nuclear Fission



- Bombardment of the radioactive nuclide with a neutron starts the process.
- Neutrons released in the transmutation strike other nuclei, causing their decay and the production of more neutrons.
- This process continues in what we call a **nuclear chain reaction**.

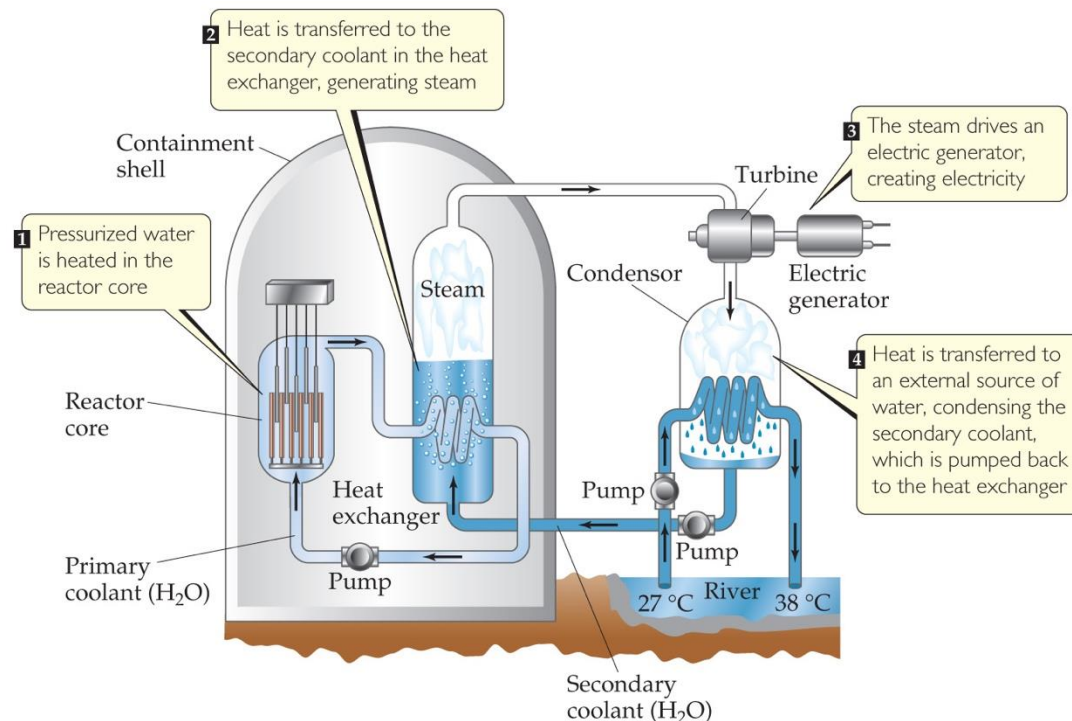
Nuclear Fission



- The **minimum mass that must be present for a chain reaction to be sustained** is called the **critical mass**.
- If **more than critical mass is present (supercritical mass)**, an **explosion will occur**. Weapons were created by causing smaller amounts to be forced together to create this mass.

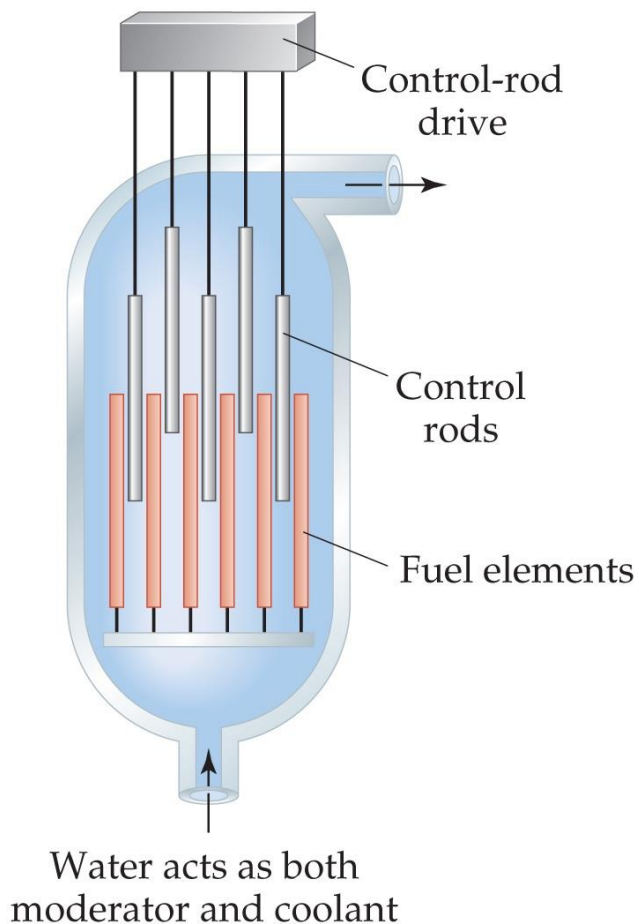
Nuclear Reactors

In nuclear reactors, the heat generated by the reaction is used to produce steam that turns a turbine connected to a generator. Otherwise, the plant is basically the same as any power plant.



Nuclear Reactors

- The reactor core consists of fuel rods, control rods, moderators, and coolant.
- The control rods block the paths of some neutrons, keeping the system from reaching a dangerous supercritical mass.

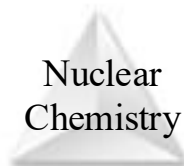
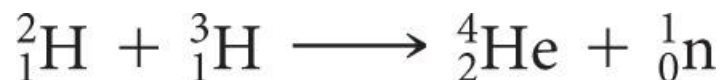


Nuclear Waste

- Reactors must be stopped periodically to replace or reprocess the nuclear fuel.
- They are stored in pools at the reactor site.
- The original intent was that this waste would then be transported to reprocessing or storage sites.
- Political opposition to storage site location and safety challenges for reprocessing have led this to be a major social problem.

Nuclear Fusion

- When small atoms are combined, much energy is released.
- If it were possible to easily produce energy by this method, it would be a preferred source of energy.
- However, extremely high temperatures and pressures are needed to cause nuclei to fuse.
- This was achieved using an atomic bomb to initiate fusion in a hydrogen bomb. Obviously, this is not an acceptable approach to producing energy.



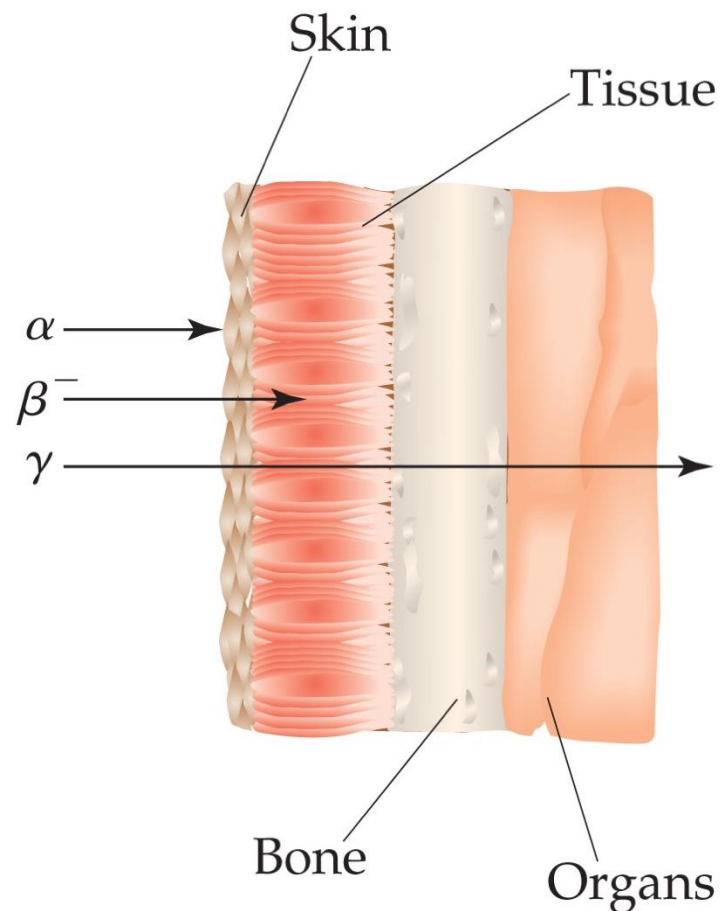
Radiation in the Environment

- We are constantly exposed to radiation.
- **Ionizing radiation** is more harmful to living systems than **nonionizing radiation**, such as radiofrequency electromagnetic radiation.
- Since most living tissue is ~70% water, ionizing radiation is that which causes water to ionize.
- This creates unstable, very reactive OH radicals, which result in much cell damage.



Damage to Cells

- The damage to cells depends on the type of radioactivity, the length of exposure, and whether the source is inside or outside the body.
- Outside the body, gamma rays are most dangerous.
- Inside the body, alpha radiation can cause most harm.



Exposure

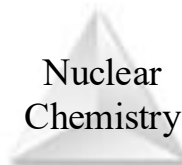
- We are constantly exposed to radiation. What amount is safe?
- Setting standards for safety is difficult.
- Low-level, long-term exposure can cause health issues.
- Damage to the growth-regulation mechanism of cells results in cancer.

Table 21.8 Average Abundances and Activities of Natural Radionuclides[†]

	Potassium-40	Rubidium-87	Thorium-232	Uranium-238
Land elemental abundance (ppm)	28,000	112	10.7	2.8
Land activity (Bq/kg)	870	102	43	35
Ocean elemental concentration (mg/L)	339	0.12	1×10^{-7}	0.0032
Ocean activity (Bq/L)	12	0.11	4×10^{-7}	0.040
Ocean sediments elemental abundance (ppm)	17,000	—	5.0	1.0
Ocean sediments activity (Bq/kg)	500	—	20	12
Human body activity (Bq)	4000	600	0.08	0.4 [‡]

[†]Data from "Ionizing Radiation Exposure of the Population of the United States," Report 93, 1987, and Report 160, 2009, National Council on Radiation Protection.

[‡]Includes lead-210 and polonium-210, daughter nuclei of uranium-238.



Radiation Dose

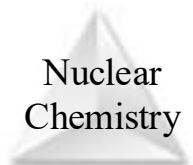
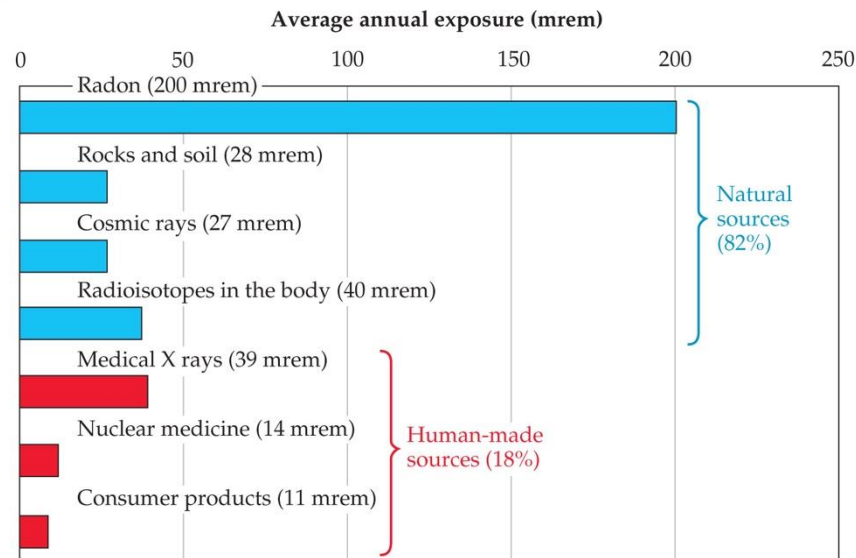
- Two units are commonly used to measure exposure to radiation:
 - **Gray (Gy)**: absorption of 1 J of energy per kg of tissue
 - **Rad (for radiation absorbed dose)**: absorption of 0.01 J of energy per kg of tissue (100 rad = 1 Gy)
- Not all forms of radiation harm tissue equally. A **relative biological effectiveness (RBE)** is used to show how much biological effect there is.
- The effective dose is called the **rem** (SI unit **Sievert**; 1 Sv = 100 rem)
- # of rem = (# of rad) (RBE)

Short-Term Exposure

- 600 rem is fatal to most humans.
- Average exposure per year is about 360 mrem.

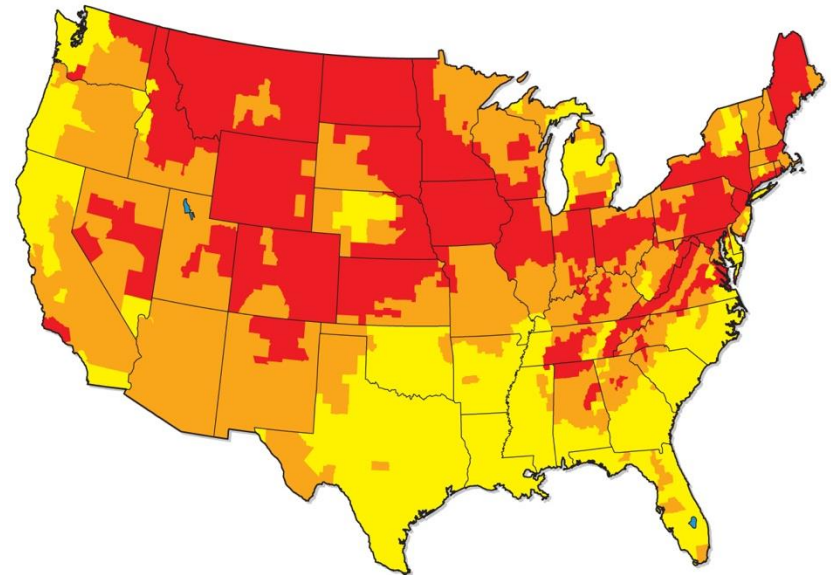
Table 21.9 Effects of Short-Term Exposures to Radiation

Dose (rem)	Effect
0–25	No detectable clinical effects
25–50	Slight, temporary decrease in white blood cell counts
100–200	Nausea; marked decrease in white blood cell counts
500	Death of half the exposed population within 30 days



Radon

- Radon-222 is a decay product of uranium-238, which is found in rock formations and soil.
- Most of the decay products of uranium remain in the soil, but radon is a gas.
- When breathed in, it can cause much harm, since it produces alpha particles, which have a high RBE.
- It is estimated to contribute to 10% of all lung cancer deaths in the United States.



- Zone 1 Predicted average indoor radon level greater than 4 pCi/L
- Zone 2 Predicted average indoor radon level between 2 and 4 pCi/L
- Zone 3 Predicted average indoor radon level less than 2 pCi/L